Image Compression via Block-wise Singular Value Decomposition

MATH/CSCI 485
Assignment #3
Gaurav Patil

March 3, 2026

1 Introduction

Images contain a large amount of data, but much of it is structured and redundant. Singular Value Decomposition (SVD) provides a powerful mathematical tool for identifying and retaining the most important structural information in a matrix while discarding less significant components.

Given a matrix $A \in R^{m \times n}$, SVD factorizes it as

$$A = U \Sigma V^T$$

where the diagonal entries of Σ (the singular values) indicate how much "energy" or importance each component contributes. By keeping only the largest k singular values, we obtain a rank-k approximation that minimizes reconstruction error in the least-squares sense.

In this assignment, SVD is applied block-wise to compress a grayscale image. Each 8×8 block is approximated using only the top-k singular values, with $k \in \{1, 2, \ldots, 8\}$. The goal is to examine how compression ratio and image quality change as k varies.

2 Methodology

2.1 Image Preprocessing

The image `Building1_img.jpg` was converted to grayscale. To allow clean partitioning into non-overlapping 8×8 blocks, the image was cropped so that both dimensions were divisible by 8. The resulting image size was:

$$224 \times 224$$

2.2 Block-wise SVD Compression

The image was divided into non-overlapping 8×8 blocks. For each block B, SVD was computed:

$$B = U\Sigma V^T$$

To compress the block, only the top-k singular values were retained:

$$B_k = U_k \Sigma_k V_k^T$$

Each reconstructed block was placed back into its original position to form the compressed image.

2.3 Compression Ratio

For an 8×8 block:

$$\text{Original storage} = 64 \text{ values}$$

$$\text{Top-}k \text{ SVD storage} = k(8 + 8 + 1)$$

Thus, the compression ratio is

$$\text{Compression Ratio} = \frac{64}{17k}$$

2.4 Reconstruction Metrics

Reconstruction accuracy was measured using the Frobenius norm:

$$\|I - I_k\|_F$$

which captures the overall difference between the original and reconstructed images.

To better reflect perceptual quality, PSNR (Peak Signal-to-Noise Ratio) was also computed:

$$PSNR = 10 \log_{10} \left(\frac{255^2}{MSE} \right)$$

Higher PSNR values indicate better reconstruction quality.

3 Results

3.1 Original Image

Figure 1: Original grayscale image

3.2 Reconstructed Images ($k = 1$ to 8)

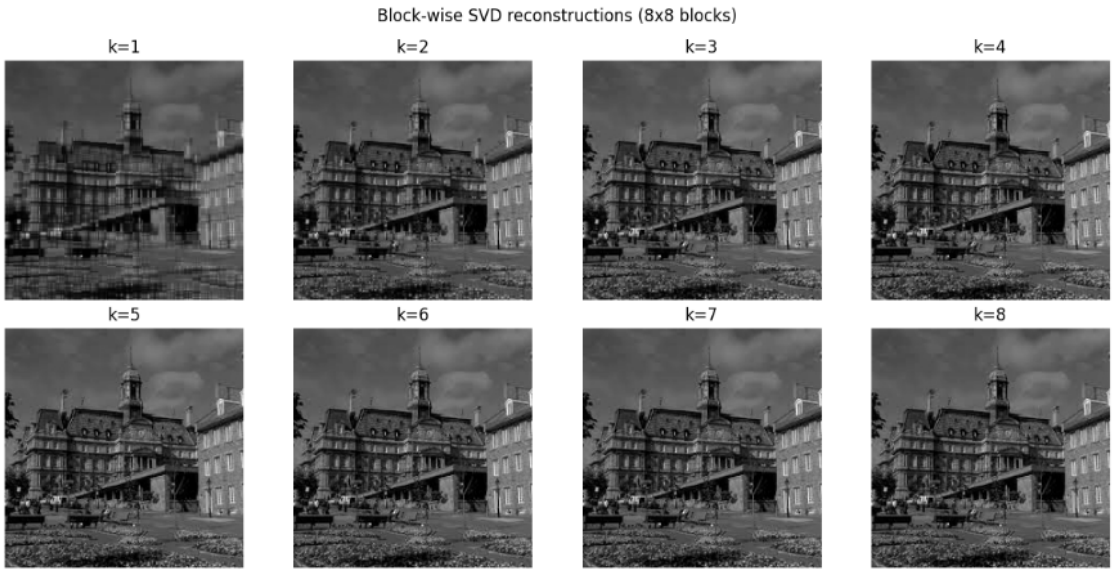

Figure 2: Block-wise SVD reconstructions for $k = 1$ through $k = 8$

3.3 Performance Plots

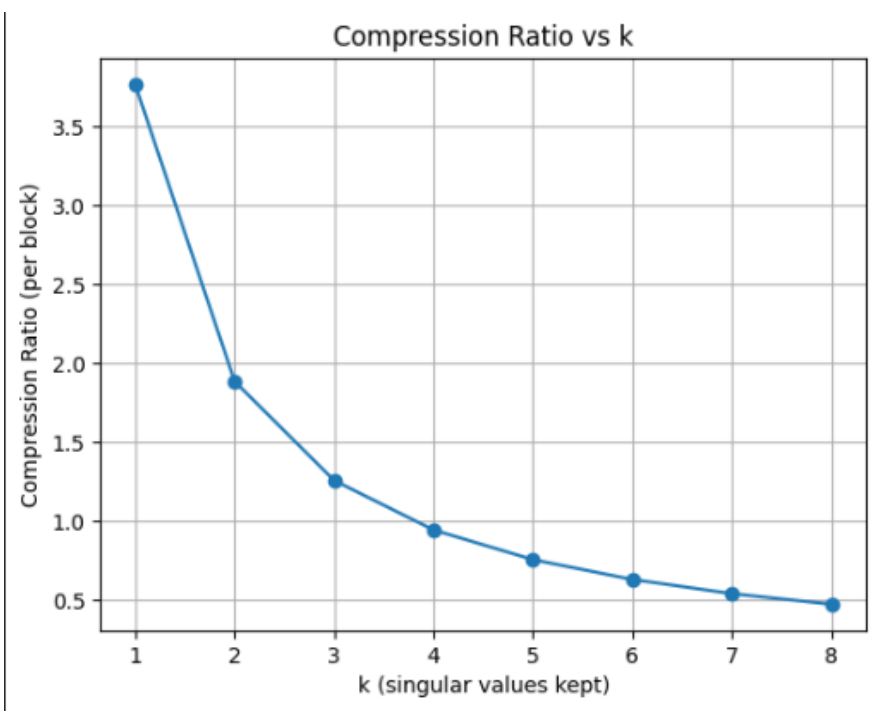

Figure 3: Compression Ratio vs k

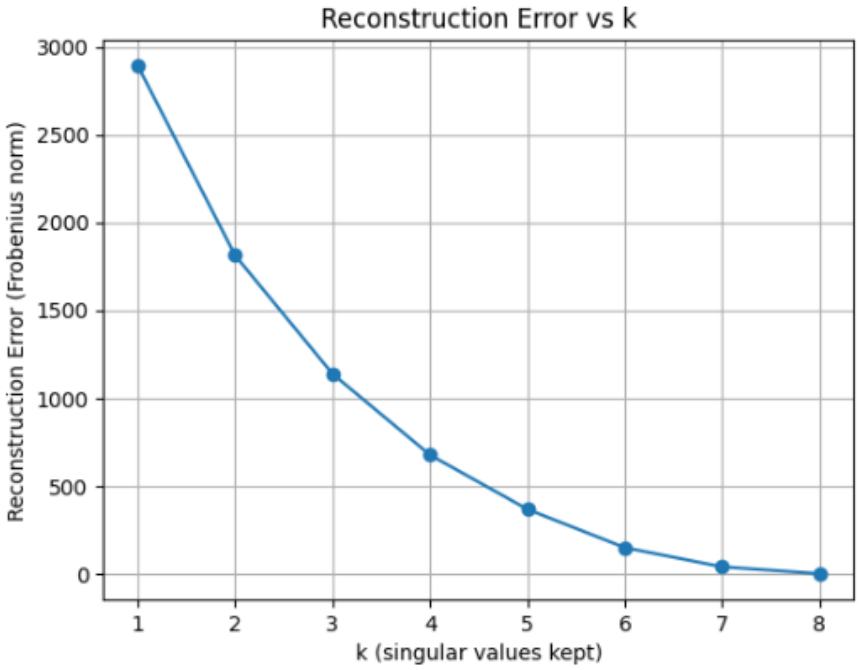

Figure 4: Frobenius Reconstruction Error vs k

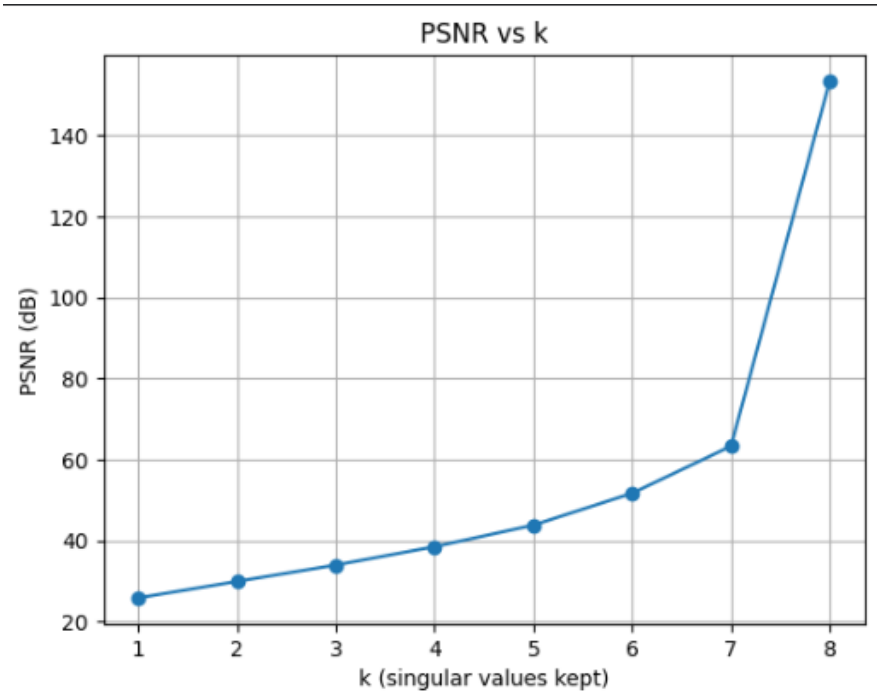

Figure 5: PSNR vs k

4 Quantitative Results

k	Compression Ratio	Frobenius Error	PSNR (dB)
1	3.7647	2893.9016	25.9061
2	1.8824	1812.6520	29.9695
3	1.2549	1137.8900	34.0138
4	0.9412	676.2787	38.5332
5	0.7529	366.8420	43.8462
6	0.6275	148.2507	51.7158
7	0.5378	38.9640	63.3225
8	0.4706	0.0012	153.3427

Table 1: Compression ratio, reconstruction error, and PSNR for each k.

5 Discussion

The results reveal a clear and intuitive pattern.

As k increases, reconstruction error steadily decreases. This is expected because keeping more singular values preserves more information. At $k = 8$, the reconstruction is nearly identical to the original image.

However, this improvement comes at a cost. The compression ratio decreases as k increases. For small k, the image is heavily compressed but loses detail, and block artifacts become visible. Edges appear blurred and fine textures are smoothed out.

As k reaches 4 or 5, most major structures are preserved. By $k = 6$ and $k = 7$, the reconstructed image appears visually very close to the original.

The PSNR values reflect this trend quantitatively, increasing rapidly as more singular values are retained.

This experiment highlights a fundamental tradeoff: retaining fewer singular values yields stronger compression but reduces visual quality, while retaining more singular values improves fidelity at the expense of storage efficiency.

6 Conclusion

Block-wise SVD provides an effective and mathematically principled method for grayscale image compression.

By adjusting the parameter k, we directly control the balance between compression and reconstruction quality. The results demonstrate how low-rank approximation captures the dominant structural information in image blocks while discarding less significant details.

This assignment illustrates how linear algebra concepts translate naturally into practical image compression techniques, and how theoretical guarantees of optimal approximation manifest clearly in experimental results.